A Constructive Introduction to Quantum Field Theory

Φ^4_3 on the torus I: the Wetterich FBSDE and renormalisation

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SISSA lectures, Lecture 5 (provisional) — notes: Lecture 5, §§12–14

Where we are

Lecture 4. Interacting field at resolution t = solution of the FBSDE

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t\left[-D\,V_T(\Phi_T)\right].$$

In the **forecast coordinate** $\psi_t = \mathbb{E}_t \Phi_T$ the forward equation has no drift and the trees are integrated exactly; the backward half is the **Wetterich equation**.

Lectures 5–7. Construct Φ^4_3 on $\mathbb{T}^3$ with this FBSDE:

1. **deterministic renormalisation**: an explicit approximate effective action $\bar{\Gamma}_q$ and its defect (today);
2. **a priori estimates** for the exact FBSDE in the coordinates of $\bar{\Gamma}_q$, uniform in the lattice (Lecture 6);
3. **removing the cutoffs** and identifying the limit (Lecture 7).

G., course notes on Φ^4_3 via FBSDEs (2026); Bailleul–Chevyrev–G. (in prep.)

What is proved

- Large mass $m^2 = q_0$ compared with the coupling: $\gamma\, q_0^{-1/2} \ll 1$, $\gamma = 3\,\lambda$. A weak dimensionless coupling on a fixed torus.
- Tightness of the lattice measures in H^{-s}, $s > \frac{1}{2}$.
- A subsequential limit: an FBSDE on $[q_0, \infty)$, driven by a Brownian motion obtained as the limit of the lattice innovations, whose terminal value has the limiting law.
- A renormalised integration by parts formula.

Not proved: uniqueness, OS axioms, infinite volume.

The guiding principle

Do not estimate Φ and F. Estimate the **error** of an explicit approximate forecast. UV stability = this error is small uniformly in the lattice.

Other constructions: Glimm–Jaffe 1973; Feldman–Osterwalder 1976; Brydges–Fröhlich–Sokal 1983; Barashkov–G. 2020; G.–Hofmanová 2021

The problem and the exact FBSDE

Lecture 5

Lattice and the mass-regulator profile

$\Lambda_N = (N^{-1}\mathbb{Z}/\mathbb{Z})^3$, $\int f = N^{-3}\sum_x f(x)$, $\Delta = \Delta_N$ nearest neighbour. Resolution parameter $q = q_0 + t^2 \in [q_0, \infty)$:

$$K_q = q - \Delta, \qquad C_q = K_q^{-1}, \qquad C = C_{q_0} = (m^2 - \Delta)^{-1}.$$

In Lecture 3 notation ($T = \infty$, and $\dot{Q} = \partial_q Q_q$):

$$Q_q = C - C_q, \quad \dot{Q} = C_q^2, \quad \dot{Q}^{1/2} = C_q, \quad S_q = C_q, \quad \mathscr{R}_q = q - q_0.$$

Free resolvents (uniform in N)

$C_q \geq 0$ as a kernel, $C_q 1 = q^{-1} 1$, $\|C_q\|_{p\to p} = q^{-1}$,

$$a_2(q) = (C_q^2)(x,x) \lesssim q^{-1/2} \wedge N^3 q^{-2}, \qquad \mathrm{Tr} C \asymp N.$$

The **bubble** $a_2(q)$ is the coincident-point variance of the shell at resolution q.

The cutoff measure

$$U(\varphi)=\frac{\lambda}{4}\int \varphi^4, \qquad V_N = U + \frac{\mu_N}{2}\int \varphi^2, \qquad \nu_N(\mathrm{d}\varphi) = \mathcal{Z}_N^{-1} \mathrm{e}^{-V_N(\varphi)} \mu^C(\mathrm{d}\varphi),$$

$\gamma = 3\lambda$, so $D^2 U(y) = \gamma y^2$ (not the counterterm γ_T of Lecture 4). The Wick ordering is now part of μ_N.

Theorem. (informal) *For q_0 large: explicit $\mu_N = -3\lambda \mathrm{Tr} C + O(\log N)$ such that (a) ν_N is tight in H^{-s}, $s > \frac{1}{2}$; (b) along a subsequence, the forecasts converge to a solution of a limiting FBSDE; (c) the limit satisfies a renormalised integration by parts formula.*

The bounds behind (a), (b) are bounds on the error Z of an approximate forecast, **uniform in N**: UV stability.

The Polchinski process as a filtering problem

Fixed N. Hidden field $\Phi_\infty \sim \nu_N$, independent Brownian motion B, observations

$$P_q = (q - q_0)\,\Phi_\infty + B_{q-q_0}, \qquad \mathscr{F}_q = \sigma(P_s : s \le q).$$

Proposition. *With* $\mathscr{K}_q(p) = \log \int \mathrm{e}^{\langle p,\varphi\rangle - \frac{q-q_0}{2}\|\varphi\|^2} \nu_N(\mathrm{d}\varphi)$*:*

- *forecast* $\psi_q = \mathbb{E}_q\,\Phi_\infty = D\,\mathscr{K}_q(P_q)$*, covariance* $\mathrm{Cov}_q = D^2\,\mathscr{K}_q(P_q) > 0$*;*
- *innovation* $W_q = P_q - \int_{q_0}^{q} \psi_s \mathrm{d}s$ *is an* $(\mathscr{F}_q)$*-Brownian motion;*
- $\mathrm{d}\psi_q = \mathrm{Cov}_q\,\mathrm{d}W_q$*, and* $\partial_q \mathscr{K} + \frac{1}{2}\mathrm{Tr} D^2\mathscr{K} + \frac{1}{2}\|D\mathscr{K}\|^2 = 0$*.*

Proof: Bayes with likelihood $\exp\{\langle P_q, \varphi\rangle - \frac{q-q_0}{2}\|\varphi\|^2\}$; Lévy's characterisation; Itô on $D\,\mathscr{K}_q(P_q)$.

$\Phi_q = C_q P_q$ is the Polchinski field

Proposition. *$\Phi_q := C_q P_q$ and $F_q := K_q(\psi_q - \Phi_q)$ satisfy*

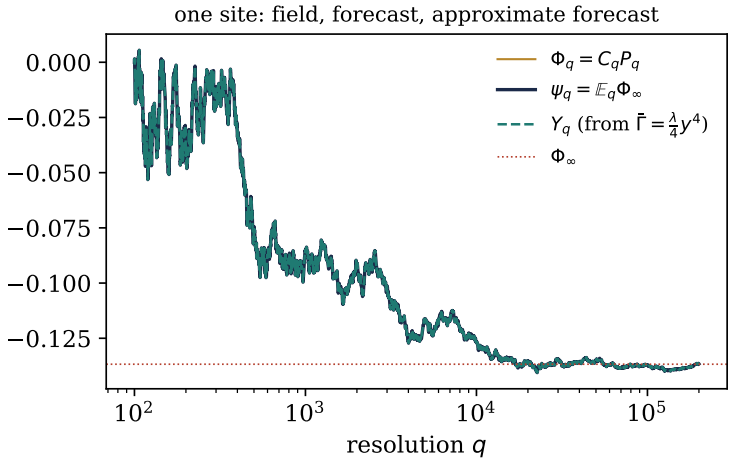
$$\mathrm{d}\Phi_q = C_q^2 F_q \,\mathrm{d}q + C_q \,\mathrm{d}W_q, \qquad F_q = \mathbb{E}_q\big[-D V_N(\Phi_\infty)\big], \qquad \psi_q = \Phi_q + C_q F_q,$$

i.e. the FBSDE of Lecture 4 for the mass profile. Cov_q *is the response* $H_q^{\mathrm{ex}}(\psi_q)$.

- Drift: $\mathrm{d}(C_q P_q) = -C_q^2 P_q \mathrm{d}q + C_q(\psi_q \mathrm{d}q + \mathrm{d}W_q)$.
- Force: posterior integration by parts, $p = K_q D\mathcal{K}_q(p) + \mathbb{E}_{\mathrm{post}}[D V_N]$ (posterior with source p), at $p = K_q \Phi_q$.
- Same law as Lecture 4: both are the Gaussian law tilted by $\mathrm{e}^{-V_N}/\mathcal{Z}_N$ of the terminal field; $\sigma(P) = \sigma(\Phi)$.

At fixed N the FBSDE is a theorem, and W is an innovation.

One site



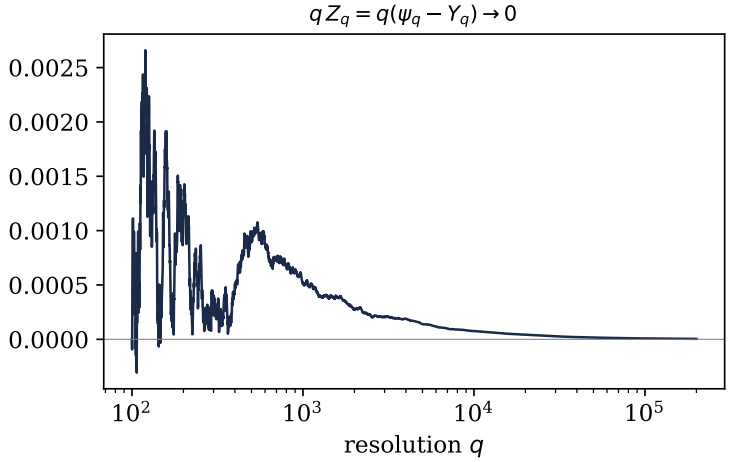


Illustration: $\nu \propto \mathrm{e}^{-\lambda\varphi^4/4}\mu^{1/q_0}$ on one site, $\lambda = 2$, $q_0 = 100$ ($g_{q_0} = 0.6$). $\Phi_q = P_q/q$, the posterior mean ψ_q, and the approximate forecast Y_q from $\bar{\Gamma} = \frac{\lambda}{4} y^4$. All converge to the hidden Φ_∞, and $q Z_q \to 0$.

Dictionary

Lecture 4	filtering at resolution q
field Φ_q	$C_q P_q$: posterior mean under the Gaussian prior μ^C
forecast ψ_q	posterior mean $\mathbb{E}_q \Phi_\infty$
response $H_q^{\mathrm{ex}}(\psi_q) = \mathrm{Cov}_q$	posterior covariance
force $F_q = K_q(\psi_q - \Phi_q)$	posterior mean of $-D V_N(\Phi_\infty)$
Brownian motion W	innovation $P_q - \int \psi$
unresolved covariance $S_q = C_q$	inverse of prior + observation precision

The lattice is the only UV cutoff: terminal resolution $q = \infty$, and $\Phi_q \to \Phi_\infty$.

The Wetterich formulation

$$\Psi_q = \mathcal{K}_q^*, \quad \Gamma_q(y) = \Psi_q(y) - \tfrac{1}{2}\langle y, K_q y\rangle, \quad H_q^{\mathrm{ex}}(y) = (K_q + D^2\Gamma_q(y))^{-1}.$$

- Up to a function of q, Γ_q is the Γ_t of Lecture 4; $\Gamma_q + \frac{1}{2}\langle y, K_{q_0} y\rangle$ is the effective average action with regulator $\mathcal{R}_q = q - q_0$.
- Coordinates: $\Phi_q = \psi_q + C_q D\Gamma_q(\psi_q)$ (Lecture 4, $x = \psi + SF(\psi)$).

Proposition. (Wetterich with a mass regulator)

$$\partial_q \Gamma_q(y) = \tfrac{1}{2}\mathrm{Tr} H_q^{\mathrm{ex}}(y), \qquad \mathrm{Cov}_q = H_q^{\mathrm{ex}}(\psi_q).$$

An **identity about the unknown ν_N**, not a construction. Strategy: an approximate $\bar{\Gamma}$ gives approximate forecast coordinates; the error should be driven by the failure of $\bar{\Gamma}$ to solve Wetterich.

Approximate forecast and error

$\bar\Gamma_q$ even, $\bar\Psi_q = \frac{1}{2}\langle y, K_q y\rangle + \bar\Gamma_q$ uniformly convex, $\bar{\mathscr{K}}_q = \bar\Psi_q^*$.

$$H_q(y) = (K_q + D^2\bar\Gamma_q(y))^{-1}, \quad e_q(y) = \partial_q\bar\Gamma_q(y) - \tfrac{1}{2}\mathrm{Tr}[H_q(y) - H_q(0)], \quad b_q = -H_q D e_q.$$

$Y_q = D\bar{\mathscr{K}}_q(P_q)$ (approximate forecast), $\quad Z_q = \psi_q - Y_q$ (error), $\quad \Xi_q = \mathrm{Cov}_q - H_q(Y_q)$.

Theorem. (exact, at fixed N)

$$\begin{aligned} \mathrm{d}Y_q &= [b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + H_q(Y_q)\,\mathrm{d}W_q, \\ \mathrm{d}Z_q &= -[b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + \Xi_q\,\mathrm{d}W_q, \end{aligned}$$

$$Y_{q_0} = Z_{q_0} = 0.$$

Proof: $\partial_q\bar{\mathscr{K}} + \frac{1}{2}\mathrm{Tr}D^2\bar{\mathscr{K}} + \frac{1}{2}\|D\bar{\mathscr{K}}\|^2 = -e\circ D\bar{\mathscr{K}} + \frac{1}{2}\mathrm{Tr}H(0)$; differentiate in p, Itô along $\mathrm{d}P = \psi\,\mathrm{d}q + \mathrm{d}W$.

Reading the system

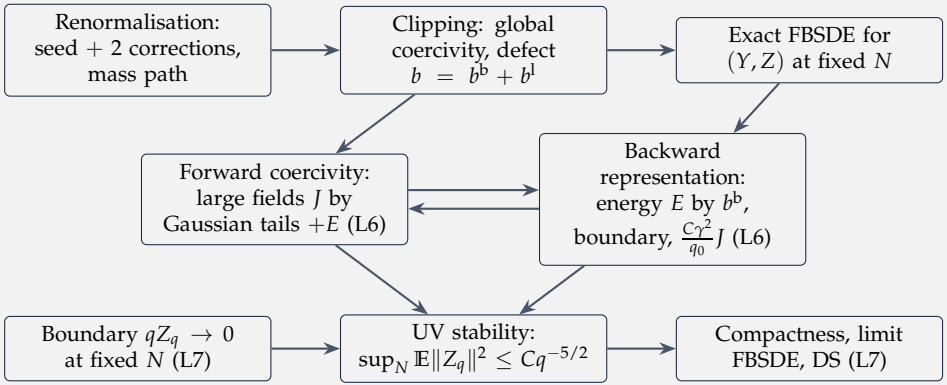
- Y: forward and explicit once Z is known; coefficients b, H built from $\bar{\Gamma}$.
- Z: backward. The terminal condition $q\, Z_q \to 0$ is **proved** at fixed N (Lecture 7), because $\bar{\Gamma}_q \to$ bare action.
- The only source of Z is the drift b, the gradient of the defect, transported by $\exp \int H$.

UV stability

The explicit action $\bar{\Gamma}$ computes the forecast up to an error Z which is small uniformly in N.

Needed: (i) a defect integrable against the backward propagator (**renormalisation**); (ii) global coercivity of $\bar{\Gamma}$ (**large fields**); (iii) a priori estimates (Lecture 6).

The structure of the construction



Renormalisation: why two of them

Lecture 5

Scales and power counting

$g_q = \gamma q^{-1/2}$ (dimensionless coupling), $\quad r_q = (q/\gamma)^{1/2}$ ($\gamma r_q^2 = q$: nonlinearity$\approx$regulator).

Free field at resolution q: $\mathrm{Var} X_q(x) \lesssim q^{1/2}$, typical size $q^{1/4} = g_q^{1/2} r_q \ll r_q$.

Rooted norm of a vertex $T(x_1, \dots, x_k)$ and field ball:

$$s(T) = \max_i \sup_{x_i} \int |T| \prod_{j \neq i} \mathrm{d}x_j, \qquad \mathsf{S}_k(f) = \sup_{\mathcal{B}_q} s\,(D^k f), \qquad \mathcal{B}_q = \{\|y\|_\infty < 2 r_q\}.$$

Natural size of the kth vertex of the **loop corrections**:

$$\mathsf{S}_k \lesssim g_q\, q\, r_q^{2-k} \qquad (k = 4 : \gamma\, g_q).$$

The loop estimate

Loop estimate

A closed loop with p vertices: $s(\text{loop}) \leq C a_2(q)\; q^{-(p-1)} \prod_i s(T_i)$ (two propagators in L^2, the others in operator norm). After the q integration a one-loop term is of size $g_q q\, r_q^{2-k}$; every further loop gains a factor g_q.

The seed and the corrections

Seed (one loop, Lecture 4, Ex. 8.7(b)): replace $D^2\Gamma$ by $D^2U=\gamma y^2$ and integrate from $q=\infty$,

$$\mathscr{L}_q(y)=\tfrac{1}{2}\mathrm{Tr}[\log\,(I+C_q^{1/2}\gamma y^2 C_q^{1/2})-C_q^{1/2}\gamma y^2 C_q^{1/2}]=\sum_{j\geq 2}\frac{(-1)^{j+1}}{2j}\mathrm{Tr}(C_q\gamma y^2)^j.$$

Quartic term = bubble, $s\,(D^4\,\mathscr{L}_q(0))=6\gamma^2 a_2(q)$.

Corrections. Π removes the constant and quadratic jets, $M_q=\mu_q+\Theta_q$ a quadratic path,

$$T_q[M,h]=\tfrac{1}{2}\mathrm{Tr}[\,(K_q+M_q+D^2\,(U+h))^{-1}-(K_q+M_q)^{-1}],$$

$$h^{[0]}=\mathscr{L},\qquad h_q^{[j+1]}=\mathscr{L}_q-\int_q^\infty \Pi\,(T_s[M,h^{[j]}]-\partial_s\,\mathscr{L}_s)\mathrm{d}s,\quad j=0,1,$$

$$\Gamma_q^{\circ}=U+\tfrac{1}{2}\langle y,M_q y\rangle+h_q^{[2]}\quad\text{on the ball.}$$

The mass

The quadratic jet is not iterated: it is solved exactly by M,

$$\partial_q M_q = -\gamma \mathrm{Tr}(R_{0,q}^2)\, I - \tfrac{1}{2} C \mathrm{tr}_{R_{0,q}^2} D^4 h_q^{[2]}(0), \qquad R_{0,q} = (K_q + M_q)^{-1},$$

with **mixed boundary conditions** $\mu_{q_0} = 0$ (physical mass) and $\Theta_q \to 0$ as $q \to \infty$ (first moment). Bare mass $\mu_N = \lim_{q\to\infty} \mu_q$: an **output** of the flow.

Proposition. (for q_0 large, uniformly in N) *Unique solution in a ball, with* $|\mu_q| \le C\gamma\sqrt{q}$, $v_1(\Theta_q) \le C\gamma^2 q^{-1/2}$; *increments* $\mathcal{N}_k(h^{[j]} - h^{[j-1]}) \le C_k g_q^{j+1} q\, r_q^{2-k}$, $4 \le k \le 10 - 2j$, $j = 1, 2$; *and the defect of* Γ°

$$\mathsf{S}_4(e_q^\circ) \le C\gamma^5 q^{-3}.$$

Ideas: trace Lipschitz estimate by loops; integrability at $q = \infty$ for $k \ge 4$; contraction for M; derivative budget $10 \to 8 \to 6$.

Two mass renormalisations

The two divergent self-energies of Lecture 4 §10.2:

- **Tadpole**: $-\gamma \int_{q_0}^{\infty} \mathrm{Tr} R_{0,q}^2 \, \mathrm{d}q \approx -3\lambda \mathrm{Tr} C \sim -N$. Wick ordering: $\frac{\lambda}{4}\varphi^4 - \frac{3}{2}\lambda c \varphi^2 = \frac{\lambda}{4} H_4(\varphi; c) + \text{const.}$
- **Sunset**: contracting the bubble in $D^4 h(0)$ with R_0^2 and integrating gives the two-loop self-energy,

$$6\lambda^2 \int C^3 \approx \frac{3\lambda^2}{8\pi^2} \log N \quad (= 2\gamma_T \text{ of Lecture 4, with } \lambda_{L3} = \lambda/4).$$

Power counting: nothing else. In the flow: the integrals from $q = \infty$ of all vertices with $k \geq 4$ legs converge; only the quadratic jet needs a boundary condition.

μ_q at fixed q has a limit as $N \to \infty$; μ_N does not. **Never interchange the two limits.**

Two correction steps

Backward propagator $\approx s/q$, so $Z_q \approx \int_q^\infty \frac{s}{q} b_s \mathrm{d}s$, and on the ball $|b_s| \lesssim s^{-1} \mathsf{S}_4(e_s)\, r_s^3$. After n steps, $\mathsf{S}_4(e_s) \lesssim g_s^{n+1} s^{1/2} r_s^{-4}$:

steps n	$\mathsf{S}_4(e_s)$	$\lvert b_s \rvert$	$\int_q^\infty \frac{s}{q} \lvert b_s \rvert \mathrm{d}s$
0	$\gamma^3 s^{-2}$	$\gamma^{3/2} s^{-3/2}$	∞ (power)
1	$\gamma^4 s^{-5/2}$	$\gamma^{5/2} s^{-2}$	∞ (log)
2	$\gamma^5 s^{-3}$	$\gamma^{7/2} s^{-5/2}$	$2\gamma^{7/2} q^{-3/2}$

Two steps are the minimum, and suffice. The defect of one step is borderline in the backward equation, as the sunset is borderline in the mass. (Heuristic threshold for Φ_3^4: $n > 1$; the general theory's sufficient criterion is $(n+2)\,\varkappa > (3+\varkappa)/2$, $\varkappa = \frac{1}{2}$, met by $n = 2$.)

The flow at large fields

Lecture 5

The large-field problem

All bounds hold on $\mathcal{B}_q = \{\|y\|_\infty < 2\, r_q\}$ only. The free field exceeds r_q with small, nonzero probability, and nothing a priori controls the interacting forecast. We need $\bar{\Gamma}_q$ **defined and uniformly convex for all fields**.

Scalar model

$\mathscr{L}_q(y) = \frac{1}{2}[\log(1+\gamma y^2/q) - \gamma y^2/q]$: power series of radius exactly r_q (singularities at $\pm \mathrm{i} r_q$). Truncations at even $j \geq 4$ are unbounded below. The exact function is harmless, $|\mathscr{L}_q''| \leq \frac{9}{8}\gamma/q$.

Perturbative control for $|y| \lesssim r_q$; outside, a non-perturbative replacement that keeps the structure the estimates use: **coercivity**, carried by the bare quartic.

One site: $q\, y + \lambda\, y^3 = p$, noise coefficient $H_q(y) = (q+\gamma y^2)^{-1} \leq q^{-1} \wedge (\gamma y^2)^{-1}$. Large fields are **more elliptic and less noisy**.

Clipping

τ smooth odd, $\tau(u)=u$ on $[-1,1]$, constant outside $[-2,2]$, $0\le\tau'\le 1$;

$$\kappa_q(y)=r_q\,\tau\,(y/r_q),\qquad \bar\Gamma_q=U+\tfrac12\langle y,M_q y\rangle+\tilde h_q,\qquad \tilde h_q=h_q^{[2]}\circ\kappa_q.$$

The quartic and quadratic parts are **not** clipped. Globally: $s\,(D^j\tilde h_q)\le C g_q q r_q^{2-j}$ ($j=2,3,4$), $s\,(D^2\partial_q\tilde h_q)\le C g_q$.

Global coercivity

For all y: $c\,(K_q+\gamma y^2)\le H_q(y)^{-1}\le C\,(K_q+\gamma y^2)$,

$$\|H_q(y)\|_{p\to p}\le\frac{1+C\,(g_q+g_q^2)}{q},\qquad \operatorname{diag}H_q(y)^2\le C a_2(q).$$

Neumann series for $H=(I+G\,(M+D^2\tilde h))^{-1}G$, $G=(K_q+\gamma y^2)^{-1}\le C_q$.

The complete defect

$b = b^{\mathrm{b}} + b^{\mathrm{l}}$, $b^{\mathrm{b}} = -H\, D\kappa\, De^{\circ}(\kappa y)$, $t_q(y) = y\, 1_{|y|>r_q}$.

Proposition. (uniformly in y, N, p)

$$\|b_q^{\mathrm{b}}\|_p \le C\gamma^{7/2} q^{-5/2}, \qquad \|b_q^{\mathrm{l}}\|_p \le C\gamma\, q^{-3/2}(\|t_q\|_p + r_q^{-1}\|y^2 1_{|y|>r_q}\|_p),$$

$$\|b_q^{\mathrm{l}}\|_2^2 \le C\gamma^4 q^{-5}\|t_q\|_6^6.$$

Sources of b^{l} (four groups): the mass term $M'y - JM'\kappa y$ ($J = D\kappa$, $M' = \partial_q M$; quadratic part unclipped, flow computed at κy); the motion of the clip; quartic vertex at y vs correction at κy; chain-rule terms with κ'', κ'''. All vanish if $\|y\|_\infty \le r_q$.

The point

b^{l} is as large as the whole drift, but **supported on large fields**. It is controlled by how often the actual dynamics visits them: $\int q^{-3/2}\gamma \|t_q(Y_q)\|_6^6\, \mathrm{d}q$ (Lecture 6).

The coordinate equation

With $DU = \frac{\gamma}{3} y^3$:

$$K_q \Phi_q = K_q (Y_q + \varrho_q(Y_q)) + \frac{\gamma}{3} Y_q^3, \qquad \varrho_q(y) = C_q (M_q y + D\tilde{h}_q(y)), \quad \|D\varrho_q\|_{p\to p} \le C(g_q + g_q^2).$$

Up to a small Lipschitz perturbation, the forecast solves the cubic elliptic equation $K_q(Y+\varrho) + \frac{\gamma}{3} Y^3 = K_q \Phi_q$ driven by the field at resolution q.

Summary of today, at every lattice:

- μ_N, M_q and a globally convex $\bar{\Gamma}_q$, all deterministic;
- the exact FBSDE for (Y, Z);
- a source whose bounded-field part is integrable (two steps), and whose large-field part is controlled by sixth powers of Y on large fields.

Next: estimates on (Y, Z) uniform in N.